THE UNIVERSITY OF BRITISH COLUMBIA

The Impact of Bilingualism on the Developmental Trajectory of Vowel Harmony Sensitivity

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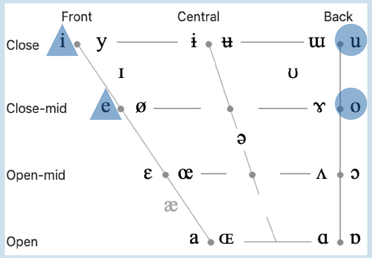
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Introduction

- **Vowel harmony (VH)** is a phonological process in which vowels within a word agree on certain phonetic feature[1]
  - e.g., backness in Turkish
    çocuk-lar "children"
    ev-ler "houses"



- Disharmonies can signal word boundaries, making VH a potential cue for word segmentation[2]
- Infants learning VH languages (e.g., Turkish[3] and Akan[4]) distinguish harmonic vs. disharmonic words by 6 months, and use VH for word segmentation by 9-11 months[5]
- Recent evidence suggests an **early, broad sensitivity to VH**:
  - 4-month-old infants primarily exposed to non-VH languages prefer harmonic compared to disharmonic sequences[6]
  - 7-month-old English-learning infants (a non-VH language) can segment speech using VH cues[7]
- This sensitivity **attenuates without relevant input**:
  - By 7-8 months, infants exposed primarily to non-VH language(s) no longer prefer harmonic words[6,7], a pattern consistent with perceptual attunement
- Bilingual infants may follow a different developmental path[8]:
  - Experience with multiple languages could maintain broader perceptual sensitivity[9]
  - However, exposure to more than one non-VH languages may also accelerate the down-weighting of VH as a cue

Does VH sensitivity decline in infants with no exposure to VH, and does exposure to more than one non-VH language change this trajectory?

Participants

- 35 mono- and 38 bilingual 6-month-old infants
- 32 mono- and 22 bilingual 10-month-old infants
- All infants had no to minimum (< 5%) exposure to VH

Methods – Central Fixation

Baseline Phase (testing preference)

- 4 fixed-length trials with a disyllabic word repeated 16 times, with 500-ms interstimulus interval (ISI)

ke-mi ***no-gu*** **vs.** ***ki-mo*** ***nu-ge***

Familiarization Phase

- 108-s continuous speech stream (45 repetitions) with no acoustic, prosodic, or statistical cues to word boundaries

de-bi-do-tu-be-pi-to-pu

Test Phase (testing segmentation)

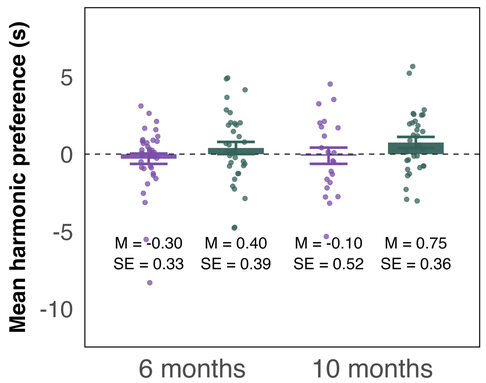
- 8 infant-controlled trials presenting a disyllabic sequence from the familiarization stream with 500-ms ISI

to-pu ***de-bi*** **vs.** ***bi-do*** ***tu-be***

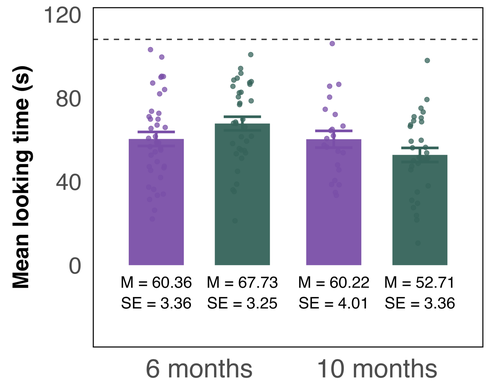
Preliminary Results

Bilinguals Monolinguals

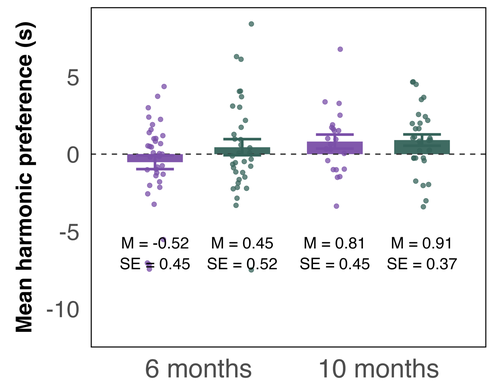
At baseline, harmony interacts with bilingualism (β = 0.18, p < .001): monolinguals preferred harmony (β = -0.22, p = .021), but no evidence for a significant preference in bilinguals (β = 0.14, p = .058).



In familiarization, 10-month-olds looked less than 6-month-olds (β = -3.63, p = .042), and age interacts with bilingualism (β = 3.83, p = .032): effect was primarily driven by monolinguals (β = -7.51, p = .002).



At test, 10-month-olds listened longer to harmonic sequences (β = 0.44, p = .021), but this effect was only significant if bilingual and monolingual infants were combined.



Summary & Discussion

- No clear evidence that infants use vowel harmony to segment words: no group reliably preferred harmonic words or disharmonic part-words at test
- English-learning monolinguals preferred harmonic word forms at baseline. A harmonic bias can therefore be present without vowel harmony input, consistent with prior work
- Bilinguals didn't show a significant preference at baseline. Exposure to two or more non-harmonic languages may accelerate attunement away from vowel harmony, providing stronger evidence that it is not a useful cue in infants' language input
- No evidence of decline in harmonic preference between 6 and 10 months in monolinguals

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